A Fast and Adaptive Coin Selection Algorithm

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Abstract

This paper introduces a novel, high-performance Coin Selection algorithm for UTXO-based blockchain systems. It is designed to consistently minimize transaction fees and reduce UTXO fragmentation while maintaining high processing speed. The approach adapts dynamically to the given set of UTXOs and targets, striking a balance between precision and computational efficiency.

1. Introduction

Blockchain networks like Bitcoin utilize Unspent Transaction Outputs (UTXOs) as the building blocks for transactions. Selecting which UTXOs to include in a new transaction significantly impacts transaction cost and network confirmation time. Efficient coin selection reduces fees, avoids wallet fragmentation, and can improve wallet performance.

Example flowchart for coin selection process

2. Problem Statement

The primary objective of coin selection is to find a subset of UTXOs whose sum meets or exceeds the required transaction amount. At the same time, it should minimize the number of inputs used, reduce the transaction fee, and produce as little change as possible.

> **Key Challenges:** Ensuring minimal transaction size, reducing change output, and maintaining high computational efficiency are critical factors in designing a practical coin selection algorithm.

3. Related Work

Several established methods address the coin selection problem:

- **Greedy Selection:** Simple and fast but often leads to suboptimal fee and change outcomes.
- **Branch and Bound (BnB):** Finds near-optimal solutions but is computationally intensive, making it unsuitable for large UTXO sets.
- **Knapsack Approach:** Theoretically sound but not practical for high-throughput environments due to complexity.

Our method seeks to maintain the speed of greedy approaches while improving cost and change outcomes through adaptive techniques.

4. Methodology

The proposed algorithm operates in four main stages:

1. **Direct Match Check:** Quickly identify if any single UTXO matches the target amount.
2. **Total Sum Check:** If the sum of all UTXOs equals the target, select all.
3. **Insufficient Funds Check:** Return an empty set if the total UTXOs value is less than the target.
4. **Adaptive Sampling:** Iteratively sample and evaluate subsets of UTXOs to find a balance between minimal inputs and low change.

Performance comparison chart

5. Evaluation

Compared to traditional methods, the adaptive coin selection algorithm offers:

- Improved fee efficiency by reducing change outputs.
- Faster execution times for large UTXO sets.
- Scalability suitable for high-throughput applications.

> **Tip:** While this algorithm excels in speed and flexibility, it is recommended to test it under various network conditions and UTXO distributions to fully understand its performance

> characteristics.

6. Conclusion and Future Work

This algorithm provides a practical approach to coin selection, balancing computational efficiency with improved transaction fee outcomes. Future enhancements could include:

- Hybrid strategies combining deterministic pruning with adaptive sampling.
- Dynamic adjustment of sampling iterations based on network congestion.
- Additional optimizations for specific wallet configurations.